

Course: BCSNT 6063 – Object Oriented Programming (2nd Semester)

Assignment Type: Practical / Tutorial

Submission: Screenshots + Program Files

Deadline: 22nd June 2026 (Monday)

University Management System

- 1) A university wants a system to store student information and display admission details.

Requirements

- Create Student class.
- Store name, roll number, and faculty.
- Use constructor for initialization.
- Use getters and setters.
- Display student profile.

- 2) University employees can be Teaching Staff or Administrative Staff.

Requirements

- Create Employee base class.
- Create Teacher and AdminStaff subclasses.
- Override displayDetails() method.

- 3) Students can register for multiple courses.

Requirements

- Create Course class.
- Create Student class.
- Maintain course list per student.
- Display registered courses.

- 4) Different students have different fee structures.

Requirements

- Create abstract class Student.
- Define abstract method calculateFee().
- Implement Undergraduate and Graduate classes.

- 5) Different departments calculate grades differently.

Requirements

- Create ResultProcessor interface.
- Implement by EngineeringDepartment and ManagementDepartment.

6) Library members can be Students or Teachers.

Requirements

- Create Member superclass.
- Extend StudentMember and TeacherMember.
- Override borrowing limits.

7) Attendance percentage calculation differs among faculties.

Requirements

- Parent class Attendance.
- Child classes EngineeringAttendance and MedicalAttendance.
- Override calculateAttendance().

8) Notifications can be sent through Email or SMS.

Requirements

- Create Notification interface.
- Implement EmailNotification and SMSNotification.

9) Every student gets a unique ID.

Requirements

- Use static counter variable.
- Generate IDs automatically.

10) Scholarship eligibility differs among programs.

Requirements

- Abstract Scholarship class.
- Child classes:
 - MeritScholarship
 - NeedBasedScholarship